

## Session 3 — Solutions and facilitation notes

Foundations of Clinical Research · Weekend 2 · Saturday · Neudata · Instructor copy

Answer keys for Practicals/s3\_exercises.md.

### Part A — Name that design

| # | Design                                   | Evidence in the abstract                                                                                           | Why (the three questions)                                                                                                                                                                                                                                                                 |
|---|------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Randomised controlled trial (RCT)</b> | "randomly allocated by a central computer"                                                                         | Assigned: yes. Random: yes. The blinded assessor and central allocation are also good practice (blinding, allocation concealment).                                                                                                                                                        |
| 2 | <b>Cross-sectional (analytical)</b>      | "On one day", "examined all 412 inpatients"                                                                        | Not assigned; comparison of mattress types; exposure and outcome measured at the same moment. We cannot tell whether the mattress came before the injury, or whether patients with injuries were moved onto special mattresses (reverse causation), which would hide part of the benefit. |
| 3 | <b>Case-control (matched)</b>            | "who developed candidaemia were compared with ... who did not", "searched the records ... before"                  | Not assigned; people selected by OUTCOME, then looked back for exposure. Only an OR can be calculated.                                                                                                                                                                                    |
| 4 | <b>Prospective cohort</b>                | "classified on arrival", "followed up ... at 90 days"                                                              | Not assigned (door-to-needle time is what happened, not what the researchers chose); grouped by exposure, then followed forward.                                                                                                                                                          |
| 5 | <b>Retrospective cohort</b>              | "divided them into those who did and did not receive" ... "then compared how many ... had delirium recorded later" | Uses old records (so many people say "case-control"), but patients were selected and grouped by EXPOSURE, and the outcome came later. Selection by exposure = cohort.                                                                                                                     |
| 6 | <b>Case series</b>                       | "We describe five patients"                                                                                        | No comparison group. Raises a hypothesis (protocol → hyponatraemia) but cannot test it.                                                                                                                                                                                                   |

| #            | Design                                                                          | Evidence in the abstract                                                   | Why (the three questions)                                                                                                                                                                                                                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7            | <b>Non-randomised intervention study</b><br>(before–after / quasi-experimental) | "before and after a ... programme, which the research team introduced"     | The investigators assigned the intervention (to everyone, at one time), but not at random. Rates may change for other reasons over time (seasonality, new staff, other initiatives) — a weak design for cause. With many monthly measurements before and after, the same project could be analysed as an interrupted time series, which is much stronger. |
| 8            | <b>Systematic review with meta-analysis</b>                                     | "searched three databases", "assessed each trial's risk of bias", "pooled" | A study of studies: systematic search + appraisal (review) + pooling of numbers (meta-analysis).                                                                                                                                                                                                                                                          |
| 9 (optional) | <b>Cluster randomised trial</b>                                                 | "wards were randomly allocated, ward by ward"                              | Assigned: yes; random: yes; but the unit of randomisation is the WARD, not the patient. Needs more patients than an individual trial, and the analysis must account for clustering.                                                                                                                                                                       |

### Facilitation notes

- Abstracts **5**, **7** and **9** cause most discussion. For 5, ask: *"How were people chosen — by what happened to them later, or by what they received?"* For 7, ask: *"Who decided the intervention? Was chance involved?"*
- Accept "cross-sectional" for abstract 2 without "analytical"; the point is the single time point.
- If a pair writes "retrospective study" alone, push for the full name: retrospective *what?*

## Part B — Choose the design for 5 questions

| # | Best feasible design                                 | Reason                                                                                  | Main weakness                                                                                                                                      |
|---|------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Cross-sectional</b><br>(point-prevalence survey)  | "How many today" = prevalence; one visit per patient, cheap and fast                    | Exposure and outcome measured together — cannot tell whether the mattress came first (patients with injuries may be moved onto special mattresses) |
| 2 | <b>Cluster randomised trial</b><br>(randomise wards) | The intervention is delivered at ward level; individual randomisation would contaminate | Needs more patients and wards; the analysis must account for clustering; few clusters give chance imbalance                                        |

| #         | Best feasible design                                     | Reason                                                                                                                     | Main weakness                                                                                          |
|-----------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 3         | <b>Interrupted time series</b> (quasi-experimental)      | The change already happened to everyone; many monthly points before and after allow comparison with the pre-existing trend | Other changes at the same time (co-interventions); no randomisation — strengthen with a control series |
| 4         | <b>Qualitative</b> — in-depth interviews or focus groups | A "why" question about experience and behaviour; numbers cannot answer it                                                  | Findings are not generalisable as numbers; depends on interviewer skill; small purposive sample        |
| 5         | Depends on the group                                     | Check it against the question type and FINER feasibility                                                                   | Must be named honestly; observational designs → "associated with"                                      |
| 6 (bonus) | <b>Case-control</b>                                      | Rare outcome; start from the few cases and compare their past drug exposures with controls                                 | Recall/record bias; choosing controls; only an OR                                                      |
| 7 (bonus) | <b>Non-inferiority RCT</b>                               | We only need to show "not unacceptably worse"; margin fixed in advance                                                     | Needs a justified margin; poor adherence can make arms look falsely similar                            |

### Facilitation notes

- The most common error is choosing an RCT for everything. For question 2, accept "RCT" only if the group explains that wards (not patients) are randomised.
- For question 3, a group that answers "before–after study" is half right — ask how to make it stronger (more time points, compare with the trend, control series).
- For the group's own question, push for **feasibility**: one unit, no funding, 3–6 months often means a retrospective cohort, cross-sectional study or ITS rather than a trial.

## Part C — Proposal builder 3: what a good answer looks like

Use this model to coach groups; do not hand it out before they have tried.

| Section                | Model answer (observational version of the running question)                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Design + justification | Retrospective cohort. Our question is about the effect of a fluid strategy on death; an RCT is the ideal design but is not feasible for one ICU without funding. Existing charts record daily fluid input/output, severity scores and outcomes, so a cohort can estimate the association between positive fluid balance and death, and generate the effect size for a future trial. We will say "associated with", not "caused". |
| Setting                | Adult mixed ICU of one teaching hospital; patients admitted over 3 calendar years; data from the electronic chart and ICU register.                                                                                                                                                                                                                                                                                              |

| Section    | Model answer (observational version of the running question)                                                                                                                                                                                                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Population | Inclusion: age $\geq 18$ ; septic shock (Sepsis-3) on ICU admission; vasopressor started within 24 h of admission; ICU stay $\geq 72$ h. Exclusion: chronic dialysis; transfer from another ICU after $> 24$ h; treatment-limitation decision within 24 h; missing fluid charts on day 1–3. |

### Common problems to catch while circulating

1. RCT chosen for a question that cannot be randomised (harm, prognosis, risk factors).
2. Exclusion criteria that repeat inclusion criteria in the negative ("age  $< 18$ ").
3. Overambitious setting (five hospitals, 2,000 patients) for a group with no funding — point them back to FINER.
4. No data source named for a retrospective design (which register, which charts?).